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Method for Ventilation Correction for Model-Based CT

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INTRODUCTION

The use of a linear approximation to measure and characterize ventilation has been previously reported [Low et al]. We propose to modify this technique for the accurate generation of HU values for model-based CT (MBCT), including 5DCT, which has the potential for replacing the qualitative and artifact-ridden 4DCT process.

AIM

To develop a novel method of ventilation correction for model-based CT (MBCT) processes that employ fast-helical free breathing CT (FHFBCT) scans. While the geometry of the image predicted by the model may be an accurate representation of the patient's anatomy, simply deforming the reference image does not correctly adjust the parenchymal HU to reflect the tissue density change caused by ventilation.

METHOD

- Ten patients underwent a clinical MBCT study that included a series of 25 fast-helical free-breathing CT scans with a simultaneous breathing surrogate.
- The first CT scan was designated the reference scan and the remaining scans were deformably registered to it using the dense displacement sampling (deeds) registration method.
- To modify the reference HU to the desired reconstructed image breathing amplitude HU, we assumed a linear relationship between HU and breathing amplitude and generated linear fits accordingly.
- We used the slope of that relationship, along with the difference between the reference and reconstructed image breathing amplitudes, to adjust the reference CT HUs.
- To reduce the impact of stochastic noise, the HU value for each voxel was taken as the mean of the 3x3x3-voxel cube centered at that voxel.
- The voxel-to-voxel fit quality, characterized by the root-mean square error (RMSE), was measured for these 10 scans, and correlated against tissue density (dHU/dA).
- Due to registration errors, voxels that included vessels typically had steep HU gradients and had corresponding registration-based HU changes in excess of the ventilation-based changes. These tissues were identified via a combination of the reference voxel HU and local HU gradients. The closest parenchymal voxels were located and dHU/dA of that voxel was used, scaled by the local air content difference.
- Sensitivity to registration error was analyzed by adding increasing Gaussian random noise to the deformation vector fields (DVFs) and correlating the standard deviation of the noise added against fit RMSE.

RESULTS

The RMSE values were found to largely fall within the stochastic image noise (<30 HU, table 1). Areas of greater RMSE values (30-70 HU) typically occurred near tissue interfaces, such as blood vessels or the chest wall. The slopes were consistently small in soft tissues. Figure 1 shows a typical fit for a parenchymal region of ventilation alongside a region of parenchymal tissue that shows no ventilation.

The majority of lung voxels were found to fall into either the parenchymal tissue HU range (-700 - -900) or the soft tissue HU range (~0) with shallow gradient (0-30 HU/mm). These voxels were found to have RMSE values for their fits between 13 and 25 HU – below the noise threshold. One such example is seen in Figure 2. Voxels that fall outside of these ranges were found to be more prone to registration issues (determined manually) as they represent boundaries between soft tissue and parenchyma.

The sensitivity of our linear fits to the registration errors was found to vary between soft and parenchymal tissue, likely due to the differences in difficulty registering these tissue types. Parenchymal tissue ventilation fits were found to degrade far less than soft tissue as a function of added registration errors, and the slopes were found to vary slightly less. The effect of registration error, however, was non-negligible in both. Figure 3 shows an example of three different ROIs from the same patient for this analysis.

Figure 4 shows a slope map for a region of lung, along with slope maps showing the tissues of interest using an HU mask. Additionally, a plot of all slopes along the red profile is shown. This plot indicates that each tissue type has an expected associated slope: near 0 for soft tissue, -30 - -50 HU/Breath for parenchyma, and between the two for boundary voxels.

Patient	Average RMSE (HU)
1	16.67
2	15.7
3	16.56
4	13.2
5	13.43
6	13.895
7	13.72
8	18.48
9	13.6
10	13.53

Table 1: The average root-mean-square error (RMSE) for each fit for each of the ten analyzed patients. Low RMSE values represent better fits. Noise fits with an average RMSE of approximately 20-30.

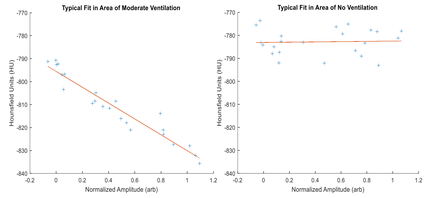


Figure 1: Typical relationship between the HU and breathing amplitudes (normalized to the 5th and 95th percentiles) for two parenchymal voxels in patient X, one with and one without measured ventilation. The lines are the linear fits used to measure the ventilation corrections.

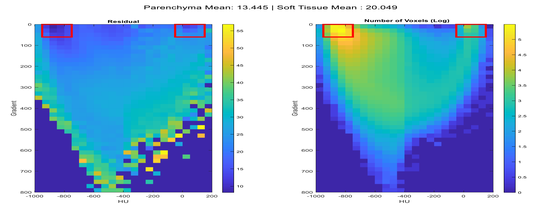


Figure 2: (a) Histogram of the mean fit residuals as a function of image gradient and HU values for the reference image of patient 10 throughout the entire lungs. (b) Histogram of the log of the number of voxels corresponding to the bins in (a). The red boxes bound the regions of high voxel density – low gradient parenchymal and soft tissue – and indicates that the RMSE for those voxels is generally low.

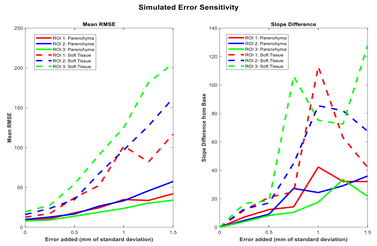


***Figure 3 (Above):** The sensitivity of the linear fits to registration error. The sensitivity was divided into two categories: parenchymal tissue sensitivity and soft tissue sensitivity. These two tissue types represent the majority of the voxels in the lung. Three different ROIs were taken in the same patient to generate the plots. One plot shows how the RMSE of the fits was affected by the added registration error (left). The other shows how the slope changed (right) by subtracting each error-added slope from the slope attained before adding error.*

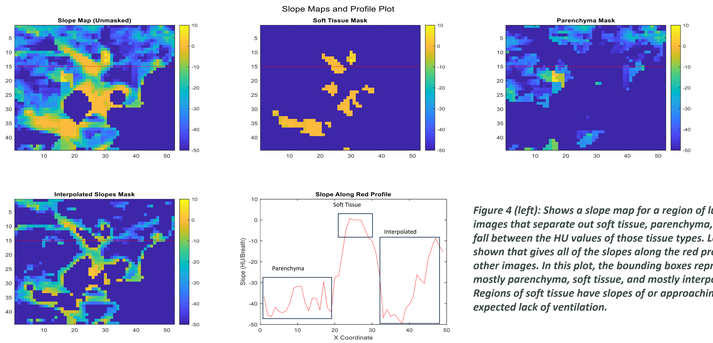


Figure 4 (left): Shows a slope map for a region of lung. Includes images that separate out soft tissue, parenchyma, and voxels that fall between the HU values of those tissue types. Lastly, a plot is shown that gives all of the slopes along the red profile drawn in the other images. In this plot, the bounding boxes represent areas of mostly parenchyma, soft tissue, and mostly interpolated values. Regions of soft tissue have slopes of or approaching 0, indicated the expected lack of ventilation.

CONCLUSIONS

The low RMSE values for the tested patients indicated that the linear approximation was suitable for measuring and accounting for ventilation-based changes in lung tissue HUs.

A large proportion of the lung voxels exhibited either parenchymal HU values with shallow gradients or soft tissue HU values with shallow gradients. In both cases, the RMSE of the fits was low, indicating suitable fits.

Parenchymal tissue proved more robust to registration error than did soft tissue and tolerably robust in general; however, registration will undoubtedly be one of the limiting factors in an approach such as this.

Further studies will determine the impact that this improved model accuracy will have on treatment planning.

REFERENCES

O'Connell, Dylan P et al. "Comparison of breathing gated CT images generated using a 5DCT technique and a commercial clinical protocol in a porcine model." *Medical physics* vol. 42,7 (2015): 4033-42. doi:10.1118/1.4922201

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CONTACT INFORMATION

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